

SAMI CHEMLAL

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Education

CentraleSupélec

2023–2027

Diplôme d'Ingénieur (MSc-equivalent), **Applied Mathematics, Machine Learning & Statistics (MDS-SDI)** GPA: **3.92/4.00**

Selected coursework: Modern Deep Learning, Transformers, NLP, Diffusion & Generative Models, Representation Learning, Transfer Learning, Reinforcement Learning, Statistical Learning, Optimization, Optimal Transport.

Lycée du Parc – CPGE (MPSI–PSI*)

2020–2023

GPA: **4.00/4.00**

Ranked **4th** in the CentraleSupélec International Entrance Examination

Research Interests

Scientific Machine Learning, Optimization & Optimal Transport, Generative Modeling, Representation Learning, Learning for Physical & Dynamical Systems, Multitask & Transfer Learning, Uncertainty Quantification.

Publications & Preprints

- T. Germain, **S. Chemlal**, R. Flamary, V. R. Kostic, K. Lounici. *Geometric Dictionary Learning of Dynamical Systems with Optimal Transport*. arXiv:2605.18276. **Submitted to ICLR 2027, main conference track.**
- **S. Chemlal**, T. Germain, R. Flamary, V. R. Kostic, K. Lounici. *Multi-Task Learning of Conditional Mean Operators: Applications to Dynamical Systems and Uncertainty Quantification*. Preprint forthcoming. **Submitted to ICLR 2027, main conference track.**

Research Experience

Research Intern – CMAP, École Polytechnique

Mar – Aug 2026

Scientific Machine Learning (Supervisors: R. Flamary, K. Lounici, T. Germain)

Palaiseau, France

- Developed **MTL-CMO**, a multi-task operator-learning framework for populations of conditional distributions, learning shared representations across related tasks and enabling **closed-form transfer** to new distributions from limited data, with accompanying statistical guarantees.
- Co-developed **DOODL**, a geometric dictionary-learning framework for populations of dynamical systems, combining **spectral operator representations, Riemannian optimization and optimal transport** to learn compact population-level representations and estimate new systems from short trajectories.
- Applied both frameworks to **uncertainty quantification and stochastic dynamical systems**, including metastable Langevin dynamics and turbulent plasma simulations, for conditional-statistics estimation, system identification, parameter recovery and regime characterization.
- Demonstrated substantial gains in low-data regimes, including **one–two orders of magnitude lower operator-estimation error** for dictionary-based transfer, and produced **two manuscripts submitted to the ICLR 2027 main conference track** during a six-month internship.

Selected Research Projects

Predictive Corruption Geometry for Generative Modeling of Physical Fields

2026

Independent Research Project Generative Modeling, Spectral Methods, PyTorch, GPU/SLURM

- Investigated **corruption geometry as an inductive bias** in generative models for physical fields, studying how the forward process controls the preservation of spatial, spectral and statistical structure during denoising.
- Derived a **mode-wise Fourier analysis of denoising risk** and analytical criteria predicting when **PDE-induced and covariance-aligned corruptions** carry complementary information; used this analysis to construct a dual-geometry representation combining physically distinct corruption processes.
- Evaluated the framework on **Kolmogorov turbulence, decaying 2D turbulence and Cahn–Hilliard dynamics**. The analytical selector recovered the choice obtained through exhaustive validation and improved reconstruction MSE relative to a tuned spectral baseline.

Robust Pattern Classification for Environmental Time Series

2025

CentraleSupélec / L2S Laboratory Deep Learning, Time Series, Robustness

Paris, France

- Formulated environmental pattern recognition as a controlled **robust time-series classification problem**, constructing synthetic benchmarks with parametrized morphology, noise, missing segments and scale variations to isolate the effect of different distribution shifts.
- Systematically compared **ANN, CNN, LSTM and Transformer architectures** under clean and corrupted-data regimes, studying robustness to severe noise, partial observations and changes in pattern geometry rather than evaluating only in-distribution classification accuracy.
- Developed an **online sliding-window inference pipeline** for detecting and classifying motifs in continuous signals, including partially observed and overlapping patterns, with the objective of transferring models from controlled synthetic benchmarks to environmental-monitoring data.

QRT Data Challenge – Cross-Sectional Return Prediction

2026

Machine Learning for Financial Time Series

- Studied **next-day cross-sectional stock-return prediction** in a noisy, low-signal setting, constructing momentum, volatility, liquidity and relative-value features from historical market data.
- Compared and ensembled tree-based learning methods to control overfitting and exploit interactions between market signals.
- Ranked **36th out of 1,447 participants (top 2.5%)** in the QRT Data Challenge.

Teaching

AI Olympiad Coach (Volunteer)

2025 – Present

AOAI – Algerian National Selection

Algeria / Remote

- Teach probability, statistical learning, optimization and deep learning for national AI-olympiad preparation; students obtained **2 gold and 2 silver medals** at the African Olympiad in AI, a European bronze medal and an **IOAI silver medal**.

Mathematics Tutor

2023 – Present

CPGE & Undergraduate Mathematics

Paris, France

- Provided individual mathematics tutoring to **CPGE and L3-level students**; students subsequently entered **CentraleSupélec, Télécom Paris, and HEC Paris**.
- Covered advanced undergraduate and preparatory-class mathematics, including **real analysis, topology, measure and integration, differential calculus, linear algebra, reduction theory, and Euclidean spaces**.

Awards & Honors

CentraleSupélec

Ranked **4th** in the international entrance examination

Chemistry Olympiad Silver Medal / 2nd place, AEFE International Chemistry Olympiad (2020)

Industry Experience

Quantitative Research Intern – Front Office

Aug 2025 – Feb 2026

Crédit Agricole CIB

Paris, France

- Developed statistical models for **credit-event and default prediction**, using linear, logistic and regularized regression with model selection, calibration and out-of-sample validation.
- Built **Monte Carlo models of correlated default events**, representing dependence through covariance structures and Cholesky factorization for portfolio-level risk simulation.
- Estimated repayment and prepayment distributions and performed statistical diagnostics using KS, Chi-squared, T- and F-tests, heteroscedasticity analysis, multicollinearity diagnostics and weighted least squares.

Technical Skills

Mathematics

Probability, Stochastic Calculus, Harmonic Analysis, Optimal Transport, Convex & Non-convex Optimization, Dynamical Systems, Operator Learning

Machine Learning

Scientific ML, Transformers, Diffusion & Flow Models, Representation Learning, Multitask & Transfer Learning, Reinforcement Learning, NLP

Programming

Python, C#, R, MATLAB, SQL; PyTorch, JAX, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, POT; CUDA/GPU Computing, SLURM, Git

Languages

English (C1), French (C2), Kabyle (C2)